

Muhan Li

778-680-8519 | muhanli.work@gmail.com | linkedin.com/in/muhan-li | github.com/muhan-alan-li

EDUCATION

University of British Columbia

BASc. Computer Engineering – GPA: 3.8 / 4.33

Sep 2020 - May 2026

Vancouver, BC

EXPERIENCE

Software Engineer

Apera AI

Sep 2025 - Apr 2026

Vancouver, BC

- Shipped self-serve asset deployment with live status tracking (REST API, React/TypeScript), replacing a manual support-ticket workflow and cutting deployment time from ~12hrs to ~5min
- Extended Forge's Python DAG job scheduler with a new node type for self-serve asset deployment, using Docker Compose for isolated execution and scoped secrets
- Added support for new class of robots with Typescript interfaces and Redux state logic to drive potential sale
- Refactored object-sharing logic to use DynamoDB transactional writes, preventing object creation errors
- Built continuous collision detection using spatial hashing in Three.js, maintaining 60+ FPS through memoization

Software Developer

Natural Resources Canada

May 2025 - Aug 2025

Vancouver, BC

- Ported the ANSE engine from Windows only .NET to cross-platform architecture to support Linux and macOS
- Shipped a C# data transformation library to standardize conversion across inconsistent input formats
- Automated ANSE build and deployment with GitHub Actions and containerized environments

Software Developer

UBC Cloud Innovation Center

Jan 2023 - Aug 2023

Vancouver, BC

- Built a serverless video pipeline with AWS Rekognition and Lambda for video capture and facial anonymization
- Shipped an anonymous drug testing platform with React and AWS, using JWT and RBAC to protect user data
- Delivered projects for two non-profit clients, translating evolving requirements into deployable software

Software Test Engineer

Sierra Wireless Inc. (now Semtech)

May 2022 - Dec 2022

Richmond, BC

- Diagnosed modem crashes with firmware and hardware engineers, from discovery through resolution
- Triageed customer reported issues, escalating critical bugs to firmware team with reproducible test cases

PROJECTS

Anteater | RAG, WebLLM

Sep 2024 - Apr 2025

- Built an offline-first RAG pipeline for Canadian student loan information, crawling and cleaning government content, generating overlapping sentence-aligned chunks, and producing a static vector index
- Configured resource-aware model deployment with WebGPU checks, ensuring safe fallback without webLLM

Find A* | Node.js, MySQL, Prisma, Android

Sep 2023 - Nov 2023

- Implemented a friendship system with MySQL and Prisma, supporting requests, adds, and removes via REST APIs
- Developed weighted review algorithm with peer feedback + time-based-decay to prioritize recent, relevant reviews

SKILLS

Languages: TypeScript, Python, C++, C#, JavaScript, HTML/CSS, SQL

Frameworks: React, Node.js, Nest.js, GraphQL, REST APIs, .NET, Flask, Three.js, NoSQL, Sass

Tools: Agile, Claude Code, Git, Docker, AWS, Jira, Ubuntu, Terraform, Cloudwatch